Real-Time Garment Style and Colour Detection Using Confidence-Gated YOLOv8 for Smart Apparel Manufacturing

***Abstract*—Garments packed at the end of a sewing line are still counted manually, causing delayed production updates, possible counting errors, and no separate visibility of production by garment style. This paper presents a camera-based system that automatically detects, counts, and records packed garments while identifying the style and main colour. Garment styles are detected using a YOLOv8n model trained on four classes, while garment colours are extracted using foreground segmentation and colour clustering. To improve reliability in the factory environment, five acceptance checks are applied to validate detection position, size, confidence, image sharpness, and consistency across frames. Accepted detections are stored with timestamps and used to generate per-style production forecasts that account for breaks and machine downtime. The detector achieved a mean average precision of 0.958 at an intersection-over-union threshold of 0.5 on the held-out test set. The results also show that detector accuracy alone does not ensure reliable factory operation, highlighting the importance of acceptance checks for rejecting false detections. The proposed system provides an automated, per-style production counting and forecasting solution for garment packing operations.**

Keywords—apparel manufacturing, closed-set recognition, confidence thresholding, edge inference, object detection, production monitoring

I. Introduction

Automatic recognition of garment style and colour has been studied in computer vision for many years. Most of that work is built on large public clothing datasets, in which clothes are worn by people and photographed for catalogues or in the street. Such studies report accuracy on clean test sets, but rarely report what happens when the same model is placed inside a factory, where the camera sees a busy table instead of one garment, and where nobody is watching the output. Section II reviews this work and sets out the gap. Existing studies do not report the acceptance rules needed to make a fixed-label detector safe in unsupervised use, and do not link garment recognition to production planning.

This gap raised a practical question. If such a system were installed in a real factory, what problems would it actually have to solve? To answer this, a field visit was made to the packing area of an apparel factory. Garment making is one of the largest export industries in Sri Lanka. At the end of every sewing line there is a packing station, where finished garments are folded and packed, and that is the point where a factory learns how much has been produced. The speed and the quality of this count decide how early a factory can tell whether an order will ship on time.

The visit recorded how the count is taken today. The observed method has three steps. A worker counts the garments packed since the last check, about every fifteen minutes. The number is written on a paper tally sheet. The sheet is typed into a production dashboard later in the shift.

This method has three weaknesses. First, the output figure is late. It is known only in fifteen-minute blocks, and only after typing, so a slow line is noticed too late to fix and can be corrected only with overtime. Second, the number passes through two manual steps. Each step can introduce an error, and no record is kept that can be checked afterwards. Third, the tally is a single number. An order that mixes shirts, T-shirts, trousers and shorts is tracked as one total, even though these styles are sewn at different speeds and have separate quantities to complete.

Based on the problems observed during the visit, this study proposes a system for real-time garment style and colour detection. A camera above the folding table can recognise each garment as it is packed and remove all three weaknesses at once. Recognising a folded garment as one of four styles is an easy task for a modern object detector. A YOLOv8n model trained for this task reached a mean average precision (mAP) of 0.958 at an intersection-over-union (IoU) threshold of 0.5 on a held-out test set.

That score turned out to be a poor guide to real behaviour. Above a working table, the same model gave confident but wrong answers. The reasons are built into the data and the label set, and are examined in Section VI.

The proposed system improves detection reliability by applying five acceptance checks around the YOLOv8n detector before a garment is counted. The checks test the position of the detection, its size, its confidence score, the sharpness of the garment image, and whether the same result appears in more than one frame. Together the checks reject unreliable detections caused by background regions, by objects that are not garments, and by unusable frames. A detection that falls below the confidence threshold is marked uncertain and is not counted, so doubtful observations never reach the production total, while valid garments are still recorded. A threshold sweep was carried out to measure the effect of this threshold. The value chosen for the factory, 0.85, gives a deliberately cautious acceptance policy.

This paper therefore makes two contributions. The first is a working camera system that reads garment style and main colour at the packing station and turns the counts into per-style production forecasts that allow for breaks and machine downtime, described in Section III. The second is a measured account of two ways a four-class detector fails in the field, together with a set of acceptance checks that restore reliable behaviour without retraining, and an analysis showing that the usual accuracy score can measure the cost of these checks but not the benefit. These appear in Sections V and VI.

II. Literature Review

A. Clothing recognition

DeepFashion [1] and DeepFashion2 [2] are the best known benchmarks for clothing detection and retrieval. Both use catalogue and street photographs, in which clothes are worn by people rather than folded flat, and both cover a large open vocabulary of categories. The task in this paper is the opposite. Only four styles are used, the garments lie flat on a table under fixed factory lighting, and the important requirement is not fine detail but a safe decision to accept or reject each observation.

B. Fast detection on small hardware

The YOLO family [3] made single-stage detection the normal choice for real-time work, and common implementations [4] make small versions such as YOLOv8n easy to train and run. Most published deployments assume a graphics processing unit (GPU). All timings in this paper were measured on a central processing unit (CPU) only, because that is the hardware available at the factory.

C. Limits of a fixed label set

Scheirer et al. [5] showed that a classifier trained on a fixed set of labels behaves unpredictably on inputs from outside that set. Hendrycks and Gimpel [6] showed that the highest class score is a useful, though imperfect, signal that an input is unfamiliar. This is the signal a confidence threshold uses. Guo et al. [7] showed that modern networks are often overconfident, which means such a threshold must be chosen from observed failures and not read from theory. Geifman and El-Yaniv [8] treated refusing to answer as a proper option, with a trade-off between coverage and risk.

D. The gap between test scores and real use

Torralba and Efros [9] showed that datasets carry hidden biases that models absorb. Recht et al. [10] measured accuracy drops on fresh test sets drawn from the same source. Sculley et al. [11] described the practical costs that surround a deployed model rather than sitting inside it.

E. Research gap

TABLE I. POSITION OF THIS STUDY AGAINST EARLIER WORK

Earlier work	Main focus	R	P	H
[1], [2]	Clothing benchmarks	No	No	No
[3], [4]	Real-time detection	No	No	No
[5], [6], [8]	Rejecting unknown inputs	Yes	No	No
[9], [10]	Dataset and test-set bias	No	No	No
This study	Garment counting in a factory	Yes	Yes	Yes

[a.] R: offers a way to refuse an input that is not a target object. P: reports the acceptance rules used in the field, and the cost. H: shown end to end on hardware without a GPU.

Table I places this study beside earlier work. Two gaps are visible. Clothing recognition studies report accuracy on clean datasets, but do not report the acceptance rules needed to make a fixed-label detector safe when it runs without a person watching, and do not report what those rules cost. Work on rejecting unknown inputs builds sound methods, but tests those methods on academic benchmarks rather than inside a working factory pipeline. No earlier study found during this work links garment style and colour reading to production forecasting that allows for breaks and downtime, on hardware without a GPU. This paper addresses both gaps.

III. METHODOLOGY

The proposed system has three layers, shown in Fig. 1: a capture pipeline at the camera, a service layer that stores data and computes forecasts, and a dashboard for the operator. All three run on one laptop with no GPU.

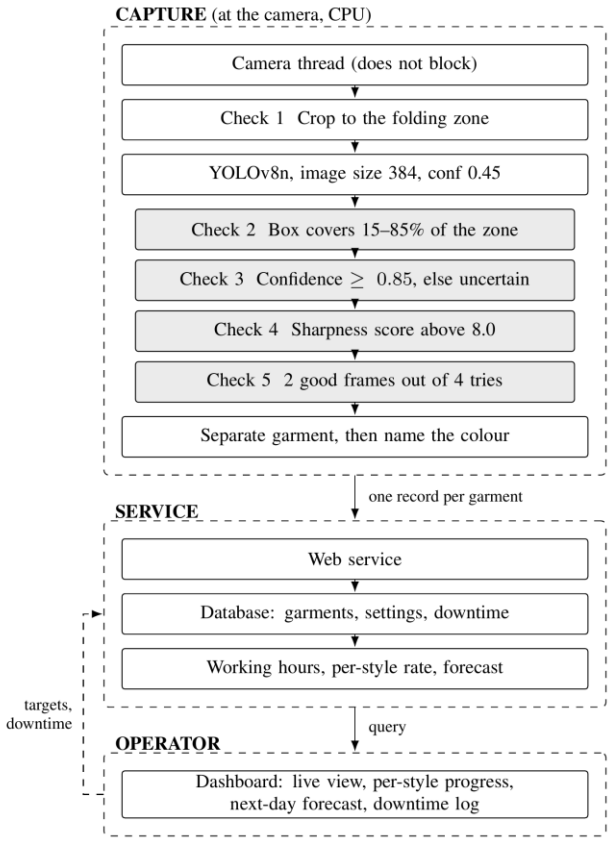


Fig. 1. The three layers of the proposed system. Check 1 is the crop; Checks 2 to 5 sit between a raw detection and a stored garment. Only Check 3 looks at the score reported by the model.

A. Capture pipeline and acceptance checks

Frames are read by a background thread so that detection never blocks the camera. Each frame is first cropped to a fixed area over the folding zone, from 0.15 to 0.85 of the frame width and 0.20 to 0.95 of the frame height. The crop is passed to YOLOv8n at an image size of 384 pixels, with a minimum score of 0.45 for a box to be drawn at all. A detection must then pass four more checks before a garment is stored.

- *Size check:* the box must cover between 15% and 85% of the cropped zone. A box that fills the whole zone is rejected.
- *Confidence check:* the score must be at least 0.85. A score between 0.45 and 0.85 is shown to the operator as uncertain instead of being dropped silently.
- *Sharpness check:* the variance of the Laplacian [15] of the garment crop must be above 8.0. This removes blurred frames caused by moving hands.
- *Agreement check:* two frames must pass all earlier checks within a window of four tries. The two frames need not be next to each other.

Every check was added after a specific failure was seen in the field, and the evidence for each is given in Section VI. After a garment is stored, the pipeline waits until the table is clear, and unlocks automatically after 90 frames so that a missed clearing cannot stop the line.

B. Reading the colour

An accepted crop is trimmed by 8% on each side, because box edges are the least accurate part of a detection, and is

resized so that the longer side is at most 320 pixels. The garment is then separated from the background in three steps.

GrabCut [12] is started from a prepared mask rather than a plain rectangle. An outer ring covering 10% of the crop is marked as probable background, a centre area trimmed by 18% is marked as certain garment, and the band between the two is left for the algorithm to decide over eight passes. A plain rectangle treats the whole crop as equally likely to be garment, which allowed hands and hair entering from the edge to survive as foreground.

Skin-coloured pixels are then removed using the YCrCb colour range of Chai and Ngan [13]. A hand resting on the cloth is the one intrusion that GrabCut most often gets wrong, because skin and beige or tan fabric can fall in the same colour group. Modelling skin directly is a long-standing solution. Xu and Sugimoto [14] used a similar idea, fitting a colour model of skin to find faces against busy backgrounds in real time. The mask is then cleaned by a morphological opening with a kernel set to 3.5% of the shorter side of the crop, which removes thin shapes such as gripping fingers without eating into the garment, and only the largest remaining region is kept.

The surviving pixels are grouped by k-means with k = 5 in the CIELAB colour space. Any group holding more than 1.5% of the garment pixels is matched to the nearest of fifteen named colours and returned in order of size. This separation is used only to read the colour. The picture shown to the operator is always the plain crop, because a mask that is good enough to steady a colour average is not always clean enough to look right.

C. Production forecasting

Every accepted garment is stored with a timestamp. Working hours for a day are the shift length minus fixed tea and lunch breaks minus any recorded downtime for machine breakdown or power failure. Output is then expressed as pieces per working hour, so that time lost to a power cut lowers the day total without spoiling the rate.

From the current rate the service reports pieces left against the target, the achieved rate against the rate needed, an efficiency percentage, a projected finish date and, when that date falls after the due date, the extra hours per day needed to catch up. Output for the next day is predicted from a five-day weighted average, with weights of 5, 4, 3, 2 and 1, combined in equal parts with a straight-line trend over the same days and scaled by the working hours planned for that day.

The same calculation runs separately for each of the four styles against its own target. This answers the third weakness found during the field visit. A single total can look on schedule while one style runs short, and the manual tally could not show this at all.

IV. Experiments

A. Hardware and software

Training and testing were carried out on a laptop with a CPU and no separate GPU. The same machine runs the deployed system. The detector uses the Ultralytics YOLOv8 implementation [4] in Python. Segmentation and colour work use OpenCV. The service layer is a Python web service with an embedded relational database.

B. Dataset

The dataset holds 3,033 images labelled across four classes: shirt, T-shirt, trouser and short. The training part holds 2,470 labelled objects: 660 shirt, 602 T-shirt, 580 trouser and 628 short, so the classes are close to balanced. The test part holds 286 images with 285 labelled objects. Almost every image contains exactly one garment. Section VI explains why this matters.

C. Training settings

YOLOv8n was trained for 50 epochs at an image size of 640 pixels, keeping the default optimiser and learning-rate schedule of the implementation. Training took 36,214 s in total, about 10.06 h, or 724 s per epoch. The smallest model in the family was chosen because the target machine has no GPU.

D. Metrics and research questions

Detection quality is reported as precision, recall and mAP at an IoU threshold of 0.5, and mAP averaged over thresholds from 0.5 to 0.95. Running cost is reported as milliseconds per frame and the resulting frame rate. Three questions are asked. RQ1 asks what accuracy the trained model reaches on the test set, and what kind of mistakes it makes. RQ2 asks whether that accuracy predicts how the same model behaves above a real folding table. RQ3 asks what the acceptance checks cost when measured by the test set itself.

V. Results

A. Detection accuracy

TABLE II. Detection Results for Each Class on the Test Set (286 Images, Confidence 0.25)

Class	Obj.	P	R	mAP@0.5	mAP@0.5:0.95
Shirt	74	0.975	0.919	0.938	0.874
Short	69	1.000	0.925	0.945	0.866
Trouser	66	0.942	0.976	0.991	0.875
T-shirt	76	0.932	1.000	0.958	0.910
All	**285**	**0.962**	**0.955**	**0.958**	**0.881**

Table II answers the first half of RQ1. The model reached a mAP of 0.958 at an IoU threshold of 0.5 across the test set, with a precision of 0.962 and a recall of 0.955. Training converged quickly. Validation mAP at an IoU of 0.5 rose from 0.720 after one epoch to 0.954 by epoch 10 and 0.969 by epoch 25, then settled near 0.985. The test score sits below that level, which is normal for a held-out set.

Two entries need care. The precision of 1.000 for short and the recall of 1.000 for T-shirt rest on 69 and 76 objects. These mean that no error of that type was seen in a small sample, not that the class is perfect.

B. Types of mistake

TABLE III. Confusion Matrix on the Test Set at Confidence 0.25. Columns Are the True Class and Rows Are the Predicted Class

Predicted	Shirt	Short	Trouser	T-shirt	Nothing
Shirt	67	–	–	–	5
Short	–	65	1	–	1
Trouser	–	3	65	–	4
T-shirt	7	1	–	76	5
Total	**74**	**69**	**66**	**76**	**15**

Table III completes the answer to RQ1 and says more than the overall score. Mistakes between classes are few and one-

sided. Seven of the 74 shirts were read as T-shirts, while none of the 76 T-shirts was read as a shirt. An overall error rate would hide this one-way bias. Trousers and shorts are mixed up mildly in both directions, three one way and one the other. No labelled object was missed completely.

The last column matters most. Fifteen boxes matched no labelled object at all: five called shirt, five T-shirt, four trouser and one short. Against 285 true objects this is a false detection rate near 5%, found on a test set that contains nothing except properly framed garments. The model therefore produces confident boxes for things that are not there, even on the kind of images it was trained on.

One note on method applies here. The same matrix produced at the default evaluation confidence of 0.001 showed a diagonal between 0.56 and 0.75 and seemed to contradict the precision and recall figures. That result came from counting very large numbers of near-zero-score boxes. The matrix is reported here at 0.25, the lowest score at which the deployed pipeline draws a box.

C. *Effect of the confidence check*

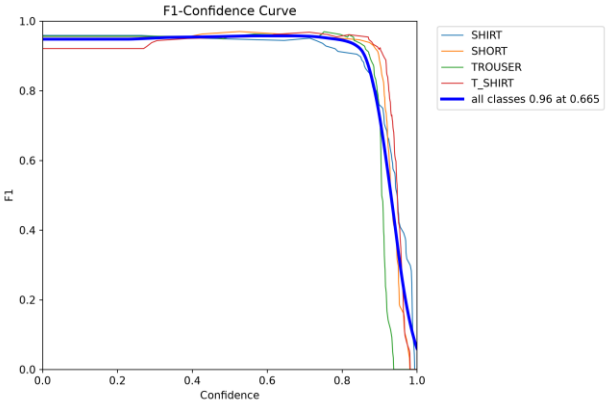


Fig. 2. F1 score against the confidence threshold on the test set. The best value of 0.96 occurs at a threshold of 0.665. The value used in the field, 0.85, sits to the right of that peak, just before the curve falls.

Fig. 2 answers RQ3 by sweeping the confidence check across its full range, which removes and restores Check 3 in a controlled way. The F1 score reaches a peak of 0.96 at a confidence of 0.665. The value used in the field, 0.85, sits well to the right of that peak, just before the curve falls away. Recall on the test set drops from 0.97 at zero confidence to about 0.90 at 0.85. Measured by the test set alone, the setting used in the field is therefore worse than the setting the test set recommends. It gives up roughly one tenth of true detections and one to two points of F1.

D. *Trial run*

A controlled trial run is reported rather than a full factory deployment. In one continuous session the assembled pipeline accepted and stored 37 garments over five and a half minutes, about one every nine seconds, with all five checks active. The stored garments were 20 T-shirt, 15 shirt, one trouser and one short. The mean confidence of accepted garments was 89.1%, and the lowest was 85.1%. Twenty of the 37 fell between 0.85 and 0.90, sixteen between 0.90 and 0.95, and one above 0.95. Main colours read were black (13), beige (8), navy (7), brown (4), grey (3) and white (2).

VI. Discussion

A. *Why the test score did not carry over*

RQ2 asks whether the score in Table II predicts behaviour above a real table. It does not, for two reasons built into the setup.

The first is that the model never learned to locate a garment. Every training image is a tight photograph of one garment filling most of the frame. The box widths and heights sit close to 1.0 in normalised units, with centres at the middle of the image. The correct answer is almost always the whole picture. A model trained on this learns to return the whole picture, and returns it for any input. Shown a wide view of a table holding several items, tools and a worker's hands, the model draws one confident box over nearly the entire frame. The test set cannot reveal this, because it comes from the same tight framing, where returning the full frame is correct. The score and the failure agree with each other. The score simply does not measure the ability the factory needs. Checks 1 and 2 handle this by recreating the training framing and rejecting boxes of an impossible size.

The second reason is the fixed label set. The model has four classes and no fifth class meaning that an object is not a garment. The four scores always add up to one, so every input is given a garment label, whether it is a cardboard box, a pair of scissors, an empty hand or a person. During testing the model labelled a person's face and hair as a shirt with 75.49% confidence. The threshold in use at that time was 0.75, so the frame was accepted and stored as a packed garment. A counter that quietly adds a piece when someone leans into view is worse than no counter, because the forecast built on it is confidently wrong. The 15 false boxes in Table III are the same problem measured on clean data. Every genuine garment seen in that session scored 82% or more, while the false detection sat between 75% and 78%. The threshold was raised to 0.85, which places it inside that gap.

B. *What the test set cannot measure*

Section V showed that raising the threshold to 0.85 costs about one tenth of recall on the test set. What it buys is the rejection of the 75.49% face, and that gain appears nowhere in Fig. 2. It cannot appear, because the test set holds only properly framed garments. There is no face in it, no cardboard box, no empty hand. The test set counts the full cost of the higher threshold and counts the benefit as zero. Any tuning method that used this test set would move the threshold back to 0.665 and bring the failure back with it.

This imbalance is the central point of the paper, and it applies beyond this system. When a fixed-label detector runs without a person watching, on inputs the test set does not contain, an accuracy score measures something other than reliability in the field. Reporting the first without the second is not enough. A simple practice follows: the acceptance rules, meaning the thresholds, the checks and the refusal rate, should be published beside the accuracy score.

Two points soften and sharpen this argument in turn. The measured recall cost overstates the real cost, because a garment is seen in many frames in a row and Check 5 needs only two of those frames. A rejected frame is not a rejected garment. That relief is narrow: it works only because the same object stays in view, and would not help a system that gets one look at each item. On the other hand, the safety margin is thin. Accepted scores in the trial run cluster just above the

threshold, with a mean of 89.1% and 20 of 37 below 0.90, while the false detection sat near 77%. A gap of about ten percentage points is enough at the tested setting, but would not survive a large change in lighting or camera position.

C. Comparison with earlier work

The behaviour reported here is a concrete case of the dataset bias described by Torralba and Efros [9] and the test-set sensitivity measured by Recht et al. [10], traced closely enough to be acted on. The fix used is an applied form of the refusal option formalised by Geifman and El-Yaniv [8], working on the highest-score signal studied by Hendrycks and Gimpel [6]. The addition to that work is not a new refusal rule but a measurement of what such a rule costs when the test set cannot see the inputs it guards against. Compared with the clothing recognition studies [1], [2], the system here uses a deliberately narrow label set but reports the acceptance rules and the hardware that those benchmarks leave out.

D. Limitations

The largest limitation is that the model has no reject class, so the confidence check is a patch and not a cure. An object that looks enough like a garment will still be accepted. Adding negative examples to the training data is the proper fix and is the main planned work.

The confidence threshold of 0.85 and the sharpness threshold of 8.0 were chosen from observations on one camera and one machine. Both are reported so that the method of choosing a value can be repeated, not so that the numbers can be copied.

Colour naming matches groups to a fixed list of fifteen colours. This is enough to tell production runs apart, but it will name unusual shades wrongly and knows nothing about patterns. Removing skin also has a known failure: a skin-coloured garment can lose its own pixels. This is handled by falling back to unmasked grouping when the separation collapses.

Finally, the evaluation in Section V is a trial run, not a factory study. It covers one session, one machine and one camera position. Style coverage is very uneven, with one trouser and one short, so half the label set is untested in the field. No separate manual count was taken while the system ran, so the figures that would describe reliability, namely how many presented garments were captured, how many were refused as uncertain, and how many accepted garments were wrong, cannot be given. The trial shows that the pipeline works end to end and where the accepted scores lie. No accuracy figure for the deployed system is claimed. For the same reason no saving in labour is claimed, because the manual tally being replaced was never timed or checked and gives no baseline. A trial against a separate manual count, over a full shift and with all four styles well represented, is the next step.

VII. Conclusion

A visit to an apparel factory packing area identified a manual counting method that is slow, easy to get wrong, and blind to the mix of styles in an order. A camera-based replacement was designed and built. The garment detector trained for it reached a mAP of 0.958 at an IoU threshold of 0.5 on a test set, yet could not be trusted in the place it was built for. The causes lie in the data and the label set. Training images that each show one garment filling the frame remove the need to locate anything, and a label set with no reject option forces a garment answer onto every object in view, including, in one recorded case, a human face at 75.49% confidence.

Reliable behaviour was restored not by improving the model but by placing five checks around it. These recreate the training framing, reject boxes of an impossible size, raise the score threshold into the gap between true and false detections, drop blurred frames, and require the model to repeat itself before anything is stored. The same layered idea returns in the colour stage, where a prepared mask, direct removal of skin tones and a cleaning step each remove one intrusion the previous step could not. The finished system runs at 37 fps on a laptop with no GPU and gives per-style forecasts that the manual tally could not.

The wider claim is deliberately narrow. For fixed-label detectors trained on single-object images and run without supervision, mean average precision measures something other than reliability in the field, and the acceptance rules that govern real use deserve to be reported beside the accuracy score.

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